

Understanding Abstract Algebra With The Rubik's Cube

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1 Introduction

*Note: it is useful to have a Rubik's Cube on hand, have a strong understanding of a Rubik's Cube, or use a simulator to help understand the examples used (<https://rubikscu.be/>)

Symbols and Definitions

- $\forall$: For all
- $\exists$: There exists
- $\in$: In
- $\equiv$: Equivalent to
- $\Leftrightarrow$: If and only if (I'll sometimes use "iff")
- $|n|$: The order (size) of n . This refers to both elements and groups.

Abstract Algebra, or Modern Algebra, is one of the fundamental upper-level pure mathematics courses required in a typical undergraduate mathematics curriculum. Many, like myself, find this course to be a formal introduction to the pure mathematics space and proof writing, as well as one of the first courses to leave formal applied mathematics spaces, such as calculus, linear algebra, and probability & statistics.

As such, this course becomes problematic, as mathematics is taught on a foundation of applied, tangible concepts. I personally found that many courses, such as Multivariable Calculus, were more understandable with the help of "objects," or things that could help you grasp the concepts by allowing you to visualize each topic or relate it to a previously understandable concept. This is often used in mathematics, such as graphs and word problems. However, with Abstract Algebra, not much of the course is visualizable or understandable across different contexts. Some topics, especially introductory group theory, are

still visualizable, with things such as the Dihedral Groups or the symmetries of polygons being readily understandable. We will begin with groups.

1.1 Assumptions & Notes

The target audience is students who have decent mathematical knowledge but have not yet entered, or have just entered, proof-based courses. Number theory or Group theory knowledge is not required.

We will be working in non-trivial finite groups. Many theorems can be disproven with either trivial ($\{e\}$) or infinite groups, so assume that theorems refer to non-trivial finite groups (unless stated otherwise).

Many definitions and notations will be ad hoc and described to help explain the direct problem at hand. Some may be universal definitions/notations, some may be used to suit problems as needed. Additionally, theorems will typically not be proven, as the goal is for readers to understand rather than to create an exhaustive textbook.

This paper is not meant to be used as a textbook, and as such, many definitions are out of order or missing altogether. Theorems and definitions are stated to help the reader understand the Rubik's Cube in context.

2 Groups

What is a group? **Definition Group:** Let G be a nonempty set together with a binary operation $(*)$ with the following properties:

1. Closure: for $a, b \in G$, $a * b \in G$
2. Associativity: for $a, b, c \in G$, $(a * b) * c = a * (b * c)$
3. Identity: there is an element $e \in G$ such that $a * e = e * a = a$, $\forall a \in G$
4. Inverse: $\forall a \in G, \exists b \in G$ such that $a * b = b * a = e$

We denote a group and its operation with the notation $(G, *)$: G being group, and $*$ being its operation.

What is a group and what is not? We can use examples to examine the following properties.

1. **Closure:** Let's try the set of odd integers: $\{\dots, -3, -1, 1, 3, 5, \dots\}$, with the operation of addition:
 $3 + 5 = 8$, which is not odd. Thus, this set is not closed.

2. **Associativity:** Let's try the set of all real numbers $\mathbb{R}$ with the operation of subtraction (Denoted $(\mathbb{R}, -)$):
 $(5 - 3) - 1 = 1$ and $5 - (3 - 1) = 3$, thus $\mathbb{R}$ is not associative
3. **Identity:** Let's try the set of all positive integers, $\mathbb{Z}^+$, with the operation of addition (Denoted $(\mathbb{Z}^+, +)$):
The additive identity is 0, which is not contained in $\mathbb{Z}^+$. So $\mathbb{Z}^+$ does not contain an identity.
4. **Inverses:** Let's try the set of all integers, $\mathbb{Z}$, with the operation of multiplication (Denoted $(\mathbb{Z}, *)$):
 $2^{-1} = \frac{1}{2}$, and $\frac{1}{2}$ is not an integer. Thus $\mathbb{Z}$ is not closed under inverses.

Thus, we have produced counterexamples for each property. Now, let's show some examples of groups:

1. $\mathbb{Z}_n$ **Groups:** the set of n integers : $\{0, 1, 2, \dots, n - 1\}$
Example: $\mathbb{Z}_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
2. $U(n)$ **Groups:** the set of integers coprime to n : $\{x \in \mathbb{Z} \mid \gcd(x, n) = 1, x < n\}$
Example: $U(21) = \{1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20\}$

But you may ask, couldn't a counterexample be shown for $\mathbb{Z}_n$ Groups? For $\mathbb{Z}_{10}$, couldn't any sum of elements past 9 disprove closure? Actually, no. This is yet another important concept in introductory Abstract Algebra: modular arithmetic.

2.0.1 Modular Arithmetic

Modular arithmetic can be thought of as "clock math," or "wrap-around math." Like the numbers on a clock, the numbers within modular arithmetic are bound to the defined space. A standard 12-hour clock can only represent the time with the given numbers. For example, 1 A.M. technically represents 13 A.M.; however, it is reduced to 1 A.M. by the clock's limitations.

Another way to think of modular arithmetic is the remainder obtained when dividing by the defined modulus. Let's use our previous group $\mathbb{Z}_{10}$. We can start with the first integer not defined in $\mathbb{Z}_{10}$: 10.

$10/10 = 1$, with remainder term 0. Thus, $10 \equiv 0 \pmod{10}$
 $11/10 = 1$, with remainder term 1. Thus, $11 \equiv 1 \pmod{10}$
The rest can be computed, with the above notation "mod 10."

What happens when we reach 20? Since $20/10 = 2$ with remainder term 0, $20 \equiv 0 \pmod{10}$. There is no difference between 10 and 20, or 30 and 40,

and so on, and every single integer can be represented by the numbers 0-9 in $\mathbb{Z}_{10}$.

Now let's look at $U(21)$ and compute it with modular arithmetic. Notice that the multiplicative group is defined as elements *coprime* to n , which means that the greatest common divisor between n and an element x is 1. Thus, all elements in $U(21)$ are coprime to 21 and have the same modular arithmetic modulo 21.

2.0.2 Group Order

The **order of a group** is the number of elements within a group. This is denoted by $|G|$. $\mathbb{Z}_{10}$ has ten elements $\{0, 1, \dots, 9\}$, so $|\mathbb{Z}_{10}| = 10$.

The **order of an element** g in group G is the smallest positive integer n such that $g^n = e$ (or $ng = 0$ in additive notation). This is denoted by $|g|$.

Note that additive groups are calculated by multiplication. Taking $\mathbb{Z}_{10}$, the order of an element is calculated as $a, 2a, 3a, \dots$ until identity 0 is obtained. For example, 1 is calculated as $1 * 1 = 1, 2 * 1 = 2, \dots, 10 * 1 = 10 \equiv 0$. Thus, 1 has order 10 in $\mathbb{Z}_{10}$.

Multiplicative groups are iterated through powers. Instead of $a, 2a, 3a, \dots$, the order of an element is $a^1, a^2, a^3, \dots$ until the identity is obtained. Taking $U(9) = \{1, 2, 4, 5, 7, 8\}$ as an example, the order of element 2 is $2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16 \equiv 7, 2^5 = 32 \equiv 5, 2^6 = 64 \equiv 1$. Thus, 2 has order 6 in $U(9)$.

2.0.3 Generators

A **generator** of a group is an element in the group with the same order as the group. The generator g in the group G iterates over each element of G . Notice how in the above examples, $|\mathbb{Z}_{10}| = 10$, and the order of 1 is 10. So 1 is a generator of $\mathbb{Z}_{10}$. The same is true for 2 in $U(9)$, where $|U(9)| = 6$ and $|2| = 6$. In context, a group with a generator can be denoted by its generator, e.g., $\langle 6 \rangle$ for $U(9)$.

Groups may not have a generator. If a generator exists in the group, the group is called *cyclic*.

Another question arises for multiplicative groups: how are higher numbers computed? For example, in $U(9)$, 7 has powers $\{7, 49, 343, 2401, \dots\}$ and is very difficult to calculate quickly. However, since $49 \equiv 4 \pmod{9}$, 343 can be calculated by multiplying 4 by 7 and reducing the result mod 9. Since $49 \equiv 4 \pmod{9}$, $4 * 7 = 28 \equiv 1 \pmod{9}$. This is proven by $9 * 38 = 342$, so 343 is 1 mod 9.

2.1 Subgroups

If a subset H of group G is under the same operation of G , then H is a subgroup of G . Note that H is still a group and satisfies the definitions of a group.

2.2 Permutation Groups

A permutation group, in simple terms, is a set of rearrangements (permutations).

Definition: Given a set A , a permutation group is a set of permutations on the set A that forms a group under function composition. The permutation is a bijective (one-to-one and onto) function.

Suppose a set $A = \{1, 2, 3, 4\}$. Naively, this set can be rearranged using the definition, i.e. $n!$ rearrangements. You will discover that Symmetric Groups (S_n), which will be discussed later, have order $n!$.

Suppose a permutation α of A where:

$$\alpha(1) = 2, \quad \alpha(2) = 3, \quad \alpha(3) = 2, \quad \alpha(4) = 4$$

Or in other words, 1 is sent to 2, 2 is sent to 3, and so on. It is easier to show it as:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{bmatrix}$$

This is not a matrix, but a table of the actions of the permutation, which we can call "array" notation. Shown below are the actual actions:

$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 3 & 1 & 4 \end{array}$$

I typically like to read it as "1 goes to 2, 2 goes to 3, 3 goes to 1, 4 goes to itself." This will be useful for cyclic notation upcoming.

Composition of permutations: Suppose permutations σ and γ on the group $\alpha = \{1, 2, 3, 4, 5\}$, defined by:

$$\sigma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 5 & 1 \end{bmatrix}$$

and

$$\gamma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 2 & 3 \end{bmatrix}$$

Suppose $(\gamma\sigma)(1)$. Since these permutations are bijective (specifically onto), $(\gamma\sigma)(1) = (\gamma(\sigma(1)))$. Thus, $\sigma(1) = 2$, and $\gamma(2) = 4$. Thus, $\gamma\sigma(1) = 4$. The entire composition can be shown:

$$\gamma\sigma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 1 & 3 & 5 \end{bmatrix}$$

2.2.1 Cycle Notation

Cycle notation is how a permutation group is denoted. We can now ditch the "matrix" notation and use our own unique notation.

Suppose cyclic notation $(1\ 4\ 5\ 6\ 8\ 3)(2)(7)$, which is, in the above notation:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 2 & 1 & 5 & 6 & 8 & 7 & 3 \end{bmatrix}$$

Now let us take our original permutations σ and γ .

- $\sigma = (1\ 2\ 4\ 5)(3)$
- $\gamma = (1\ 5\ 3)(2\ 4)$

How are these written? We can walk through each parentheses as shown. We will always start at 1, and continue down the line until we hit a repetition or the permutation ends.

1. σ : 1 goes to 2, 2 goes to 4, 4 goes to 5, and 5 goes back to 1. Thus, we have hit a repetition, and the parentheses close, as we have completed a cycle (loop): $(1\ 2\ 4\ 5)$. There is no need to repeat numbers; a number at the end of the parentheses returns to the first number. Since 1, 2, 4, and 5 are all used, we can go to the next available number, 3. 3 goes to itself, so we write (3) .
Thus, $\sigma = (1\ 2\ 4\ 5)(3)$
2. γ : 1 goes to 5, 5 goes to 3, 3 goes to 1, and we close the parentheses: $(1\ 5\ 3)$. The next available number is 2: 2 goes to 4, 4 goes to 2, and we close the parentheses: $(2\ 4)$.
Thus, $\gamma = (1\ 5\ 3)(2\ 4)$

Note that in σ , 3 loops to itself, and is self explanatory, which can be omitted: $\sigma = (1\ 2\ 4\ 5)(3) = (1\ 2\ 4\ 5)$. The latter is usually used instead of the former, and we can call 3 "closed".

Our permutation composition $\gamma\sigma$ is:

$$\gamma\sigma = (1\ 5\ 3)(2\ 4)(5)(1\ 2\ 4\ 5)(3)$$

While this is *technically* a finished definition, it is more desirable to express it in *disjoint cycle* form, or cycles with no numbers in common (i.e., a single number does not appear in various cycles).

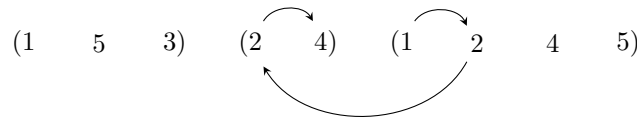
We can compute $\gamma\sigma$ with the following directions:

- Start at the smallest number in the rightmost parentheses (Does not have to be the last parentheses). Write this number down in your output.
- Perform one action per parentheses. Remember that each number "goes" to the number on its right, and the last number in each parentheses goes to the first number in the parentheses.
- Stop at the number in the leftmost parentheses possible. If this number does not exist in the left parentheses (if there are any at all), stop, and write this number down. Restart from a) with the number you wrote down.
- Continue applying the permutation until you return to the starting number. Close the parentheses, and restart at a) with the lowest number not already listed.
- Repeat a-d with the smallest unused number until all numbers have been included.

Notes:

- The number you begin with does NOT go to the number it goes to in the first action; it goes to the last available number in part c). This will become clearer in examples.
- The goal is to reduce down to disjoint cycles.

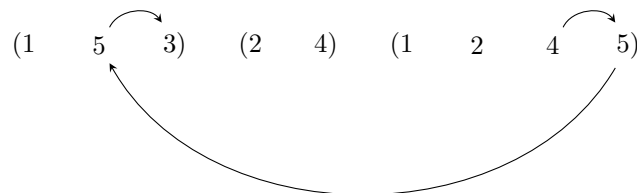
Now let's compute $\gamma\sigma$ (omitting closed parentheses):



1 goes to 2, and moving to the next parentheses: 2 goes to 4. There is no 4 to the left, so 1 goes to 4. Thus, so far we have:

$(1 \ 4$

Now we continue with the rightmost 4:



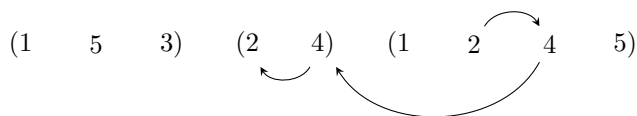
4 goes to 5, and 5 goes to 3, so 4 goes to 3:

$$(1\ 4\ 3)$$

Notice the rightmost 3 is in the leftmost parentheses, and also goes to 1, so we can close the parentheses:

$$(1\ 4\ 3)$$

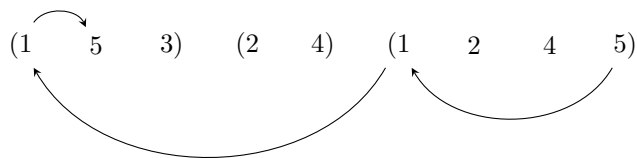
2 is the lowest unused number, so we start with 2:



2 goes to 4, and 4 goes back to 2. Thus 2 goes to itself, and is closed. We can simply write (2).

$$(1\ 4\ 3)(2), \text{ or just } (1\ 4\ 3)$$

5 is the final remaining number:



5 goes to 1, and 1 goes back to 5. Thus 5 goes to itself, and is closed. We can write (5).

$$(1\ 4\ 3)(2)(5), \text{ or just } (1\ 4\ 3)$$

We can refer back to the array notation of $\gamma\sigma$:

$$\gamma\sigma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 1 & 3 & 5 \end{bmatrix}$$

See that 1 goes to 4, 2 goes to itself (closed), 3 goes to 1, 4 goes to 3, and 5 goes to itself (closed). This verifies our permutation, and we conclude:

$$\begin{aligned} \gamma\sigma &= (1\ 5\ 3)(2\ 4)(5)(1\ 2\ 4\ 5)(3) \\ &= (1\ 4\ 3) \end{aligned}$$

Let's try some other examples!

$$1. \alpha\beta = (1\ 2)(4\ 5)(1\ 5\ 3)(2\ 4)$$

Answer:

1 to 5, 5 to 4, so 1 to 4: (1 4

4 to 2, 2 to 1, so 4 to 1: (1 4)
 2 to 4, 4 to 5, so 2 to 5: (1 4)(2 5)
 5 to 3: (1 4)(2 5 3)
 3 to 1, 1 to 2, so 3 to 1: (1 4)(2 5 3)
 $\alpha\beta = (1\ 2)(4\ 5)(1\ 5\ 3)(2\ 4) = (1\ 4)(2\ 5\ 3)$

2. $\sigma_1\sigma_2 = (1\ 3\ 2\ 4)(5\ 6\ 7)(1\ 4\ 7\ 3\ 5)(2\ 6)$
 1 to 4, 4 to 1, 1 is closed
 2 to 6, 6 to 7, so 2 to 7: (2 7)
 7 to 3, 3 to 2, so 7 to 2: (2 7)
 3 to 5, 5 to 6, so 3 to 6: (2 7)(3 6)
 6 to 2, 2 to 4, so 6 to 4: (2 7)(3 6 4)
 4 to 7, 7 to 5, so 4 to 5: (2 7)(3 6 4 5)
 Since all numbers are accounted for, this is just a formality:
 5 to 1, 1 to 3, so 5 to 3: (2 7)(3 6 4 5)
 $\sigma_1\sigma_2 = (1\ 3\ 2\ 4)(5\ 6\ 7)(1\ 4\ 7\ 3\ 5)(2\ 6) = (2\ 7)(3\ 6\ 4\ 5)$

Interestingly enough, disjoint cycles commute.

Inverse permutations: An inverse permutation is the permutation that brings another permutation to the identity. You will see that this boils down very neatly.

Definition: if $\sigma = (a_1\ a_2\ \dots\ a_n)$, then $\sigma^{-1} = (a_1\ a_{n-1}\ \dots\ a_2)$

Example: Suppose $\alpha = (1\ 2\ 3\ 4\ 5\ 6)$. Then, $\alpha^{-1} = (1\ 6\ 5\ 4\ 3\ 2)$. Let's check $\alpha\alpha^{-1}$:

$$\begin{aligned}\alpha\alpha^{-1} &= (1\ 2\ 3\ 4\ 5\ 6)(1\ 6\ 5\ 4\ 3\ 2) \\ &\quad 1\ \text{to}\ 6,\ 6\ \text{to}\ 1\ (\text{closed}) \\ &\quad 2\ \text{to}\ 1,\ 1\ \text{to}\ 2\ (\text{closed}) \\ &\quad 3\ \text{to}\ 2,\ 2\ \text{to}\ 3\ (\text{closed}) \\ &\quad 4\ \text{to}\ 3,\ 3\ \text{to}\ 4\ (\text{closed}) \\ &\quad \text{And so on.}\end{aligned}$$

Thus, $\alpha\alpha^{-1} = (1)(2)(3)(4)(5)(6) = e$.

This works in disjoint cycles too. Suppose $\alpha = (1\ 2\ 5)(3\ 4)$. Then, $\alpha^{-1} = (1\ 5\ 2)(3\ 4)$. The proof of reaching the identity e can be left as an exercise for the reader.

2.2.2 Order of Permutation Groups

The order of a permutation group is the least common multiple of the lengths of the cycles. Take $\alpha\beta = (1\ 4)(2\ 5\ 3)$ and $\sigma_1\sigma_2 = (2\ 7)(3\ 6\ 4\ 5)$ from above.

- $\alpha\beta$ has cycle lengths 2 and 3, and the least common multiple of 2 and 3 is 6. Thus, $|\alpha\beta| = 6$.

- $\sigma_1\sigma_2$ has cycle lengths 2 and 4, and the least common multiple of 2 and 4 is 4. Thus, $|\sigma_1\sigma_2| = 4$.

If a permutation group only has one cycle, then the order of the group is the length of that cycle.

2.2.3 Symmetric Group

S_n = the permutations of the set $\{1, 2, \dots, n\}$. The order of S_n is $n!$.

1. S_1 : Permutations of 1, which is just $\{e\}$. $|S_1| = 1! = 1$.
2. S_2 : Permutations of $\{1, 2\}$, where $e = (1)(2)$, $S_2 = \{e, (1\ 2)\}$
3. S_3 : $\{e, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$

Notice how S_n gets very large very quickly. S_n has 24 elements, and S_5 has 120. Thus, without writing out the whole group, we can write out the types of elements. Take S_4 :

- The identity
- 6 transpositions (like $(1\ 2)$)
- 3 double transpositions (like $(1\ 2)(3\ 4)$)
- 8 three-cycles (like $(1\ 2\ 3)$)
- 6 four-cycles (like $(1\ 2\ 3\ 4)$)

2.2.4 2-Cycles

2-Cycles are exactly as they sound: cycles of 2. For $n > 1$, every permutation in S_n can be represented as a product of 2 cycles.

Example: $(1\ 2\ 3\ 4\ 5\ 6) = (1\ 6)(1\ 5)(1\ 4)(1\ 3)(1\ 2)$

This is interpreted as:

1 goes to 2: $(1\ 2)$

2 goes to 1, 1 goes to 3: $(1\ 2\ 3)$

3 goes to 1, 1 goes to 4: $(1\ 2\ 3\ 4)$

And so on.

Hence, we are using 1 as a "dummy variable" to iterate over the permutation, since it is the first number and easy to refer to.

Other Examples:

$(1\ 5\ 7\ 3\ 2\ 4)(1\ 6\ 3) = (1\ 4)(1\ 2)(1\ 3)(1\ 7)(1\ 5)(1\ 3)(1\ 6)$

or in disjoint form:

$(1\ 6\ 2\ 4)(3\ 5\ 7) = (1\ 4)(1\ 2)(1\ 6)(3\ 7)(3\ 5)$

Odd vs. Even permutations

A permutation is odd if it has an odd number of disjoint 2-cycles, and even if it has an even number.

Even permutations end up forming a group, A_n , or the Alternating Group.

2.2.5 Alternating Group

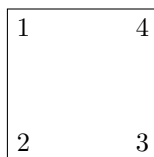
The Alternating Group A_n is the group of all even permutations of n elements. It is a subgroup of S_n . A_n leads to normal subgroups and simple groups, two of the most important topics in Abstract Algebra, with connections to further mathematics in Galois theory (named after Evariste Galois) and field theory.

- Normal Subgroups: A subgroup H of the group G is normal $\Leftrightarrow ghg^{-1} \in H \mid \forall g \in G, h \in H$. This is denoted $H \triangleleft G$
- Simple Groups: A simple group is a group whose subgroups are only the trivial subgroup and itself.
- Solvable Groups: A group is solvable if you can break it down into a chain of normal subgroups: $G_0 \triangleleft G_1 \triangleleft G_2 \dots \triangleleft G_n = G$ such that each quotient G_{i+1}/G_i is abelian. This is known as a composition series.

For $n \geq 5$, A_n is simple. A_5 is the smallest non-abelian simple group. A_5 also disproves the existence of a quintic formula for polynomials. While there is a general quadratic formula and the lesser-known (and significantly more complex) cubic and quartic formulas, Galois theory shows that there is no way to produce a general quintic formula. The general quintic has the Galois group S_5 , which contains A_5 , and, by definition of solvable groups, S_5 is not solvable because A_5 is non-abelian.

2.3 Dihedral Group Order 8 (D_4)

What is a Dihedral Group? As stated above, it is the “symmetries of polygons,” but that hardly makes sense. We will use the example of the symmetries of a square, which will be useful in our applications to the Rubik’s Cube. Symmetries are actions performed on a given shape that do not alter its original position. Keep that in mind for later, as it will become clearer with an example. Now, suppose a square with labeled corners like the one below:



The Dihedral Group is the group of all actions on the square that do not alter its shape. This is extended to all Dihedral Groups, as they are categorized (in simple terms) as:

- Having two actions: Rotations, and Reflections

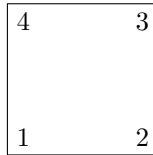
- Having $2n$ elements for a group

With examples:

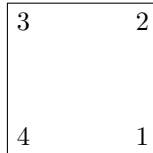
1. Symmetries of an equilateral triangle
2. Symmetries of a square
3. Symmetries of a hexagon

And so on.

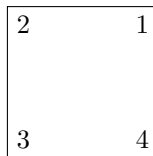
A rotation, as is pretty obvious, is a single turn of the polygon. For our examples, we will name a single rotation R (also denoted R_{90°) and categorize it as a single counterclockwise rotation of our square. Imagine the above square as the initial position, which we will label as the identity, or e . Whenever we act on our square, we will label the resulting position with the action performed. Now, after rotating the square once, we are left with the position R :



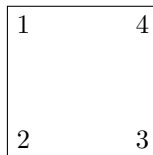
Now, we can do this again, with the notation R^2 (or R_{180°):



And again, with the notation R^3 (or R_{270°):



Some of you may have noticed that there is an interesting situation: what is the next rotation? Let's look at R^4 :



But that is the same position as the identity! It stands to reason now that $R^4 = e$

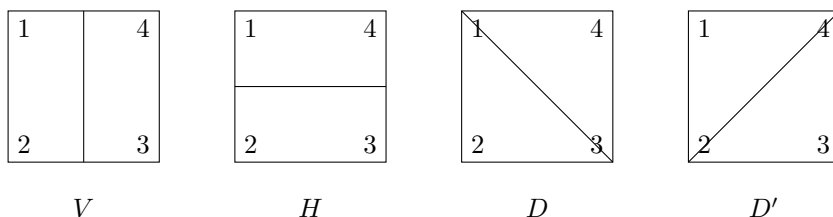
This concept, that the structure will come back in a “circle,” is extremely important to understand in Abstract Algebra, and will be useful to understand, as many group types are comprehensible once this idea is mastered (e.g., $\mathbb{Z}_n$, $U(n)$, permutation groups, etc.)

We can construct a semi-complete Cayley Table to map what we have so far. Remember the conversions, as we are working in a space only up to R^3 . With this simple subgroup of D_4 : $\{e, R, R^2, R^3\}$, we know (by one of the definitions of a subgroup) that this is closed, i.e., no combination of elements will produce an element not in the group. Thus, $R \times R^2 = R^3$, $R^2 \times R^3 = R^5 \equiv R$, and so on.

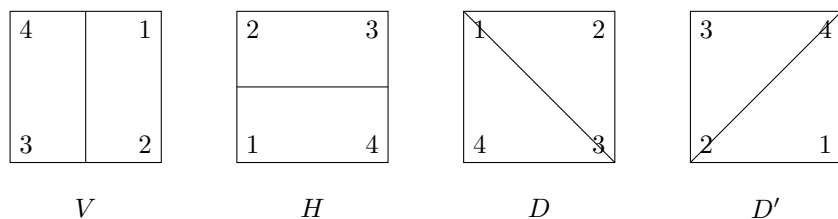
$\cdot$	e	R	R^2	R^3
e	e	R	R^2	R^3
R	R	R^2	R^3	e
R^2	R^2	R^3	e	R
R^3	R^3	e	R	R^2

Now let's move on to reflections. A reflection is defined by drawing a single line from one side or corner of the square to the opposite side and “reflecting,” or flipping, the corners across the reflection line. There are four possible reflections, defined by the line:

- V: A vertical line through the of middle square
- H: A horizontal line through the middle of the square
- D: A diagonal line from the top left corner to the bottom right corner
- D': A diagonal line from the top right corner to the bottom left

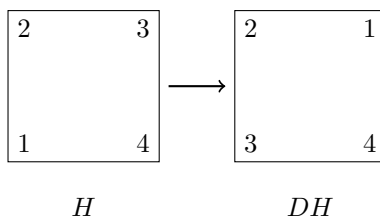


Reflections are simply flipping the square along the line, which results in “switching” of corners. As such, V, H, D, and D' are as follows:

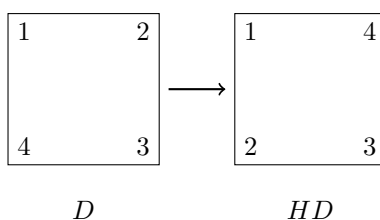


We have completed all possible positions of a square! Here, some simple computation is required. Note another important distinction used in Abstract Algebra: abelian vs. non-abelian groups. Abelian, in simple terms, is commutativity, or $ab = ba$. Many groups are abelian, especially $\mathbb{Z}_n$ and $U(n)$. However, we can see that D_4 is NOT abelian. The multiplication of two operations proves this in D_4 . Also note that not every combination of operations produces a non-abelian example. Notably, the rotations subgroup $\{e, R, R^2, R^3\}$ is actually abelian, which is obvious, as each operation does the same thing, just with a different magnitude.

Let's try the operations DH and HD . Remember that in group theory, multiplication is read right to left, so DH is interpreted as H applied first, then D .



Remember that $D_4 = \{e, R, R^2, R^3, V, H, D, D'\}$, or that there are no other possible positions (closed). DH is clearly not in D_4 , so and when looking at our existing positions, we notice that $DH = R^3$. Now let's try HD :



Which is e . Thus, $DH \neq HD$, and D_4 is non-abelian. As D_4 contains 8 elements, it is a group of order 8, and is defined as a non-abelian group of order 8.

Now, let's compute the rest of the Cayley Table

$\cdot$	e	R	R^2	R^3	H	V	D	D'
e	e	R	R^2	R^3	H	V	D	D'
R	R	R^2	R^3	e	D'	D	H	V
R^2	R^2	R^3	e	R	V	H	D'	D
R^3	R^3	e	R	R^2	D	D'	V	H
H	V	D	V	D'	e	R^2	R	R^3
V	H	D'	H	D	R^2	e	R^3	R
D	D	V	D'	H	R^3	R	e	R^2
D'	D'	H	D	V	R	R^3	R^2	e

Now, let's return to symmetries and how the square has helped us define them. As noted for the subgroup $\{e, R, R^2, R^3\}$, the R rotations can also be denoted as $\{e, R_{90^\circ}, R_{180^\circ}, R_{270^\circ}\}$. If given a physical "square," a 90° turn preserves the "look" of the square, or in other words, the square does not change visually (aside from the corners switching). Alternatively, the symmetries of an equilateral triangle do not include 90° turns, but 120° turns, as they preserve the position of the equilateral triangle.

3 The Rubik's Cube

Invented in 1974 by Ernő Rubik, the Rubik's Cube has become one of the most well-known combinatorial puzzles of our era. An architect by trade, Ernő Rubik initially developed the toy as a "good task for his students." And a good task it was! The cube quickly gained popularity and reached international crowds by 1980. There have been over 500 million cubes sold since.

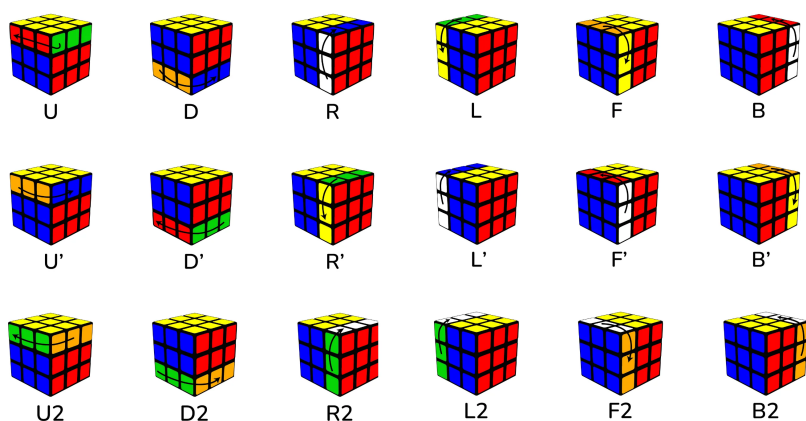
Many people, like myself, initially picked up the cube with confusion and awe. Only knowing simple tricks like the checkered pattern, I had been scared to lose the initial solved state that the cube came in from its original packaging, knowing that I would not be able to solve it. Determined to learn the Rubik's Cube through and through, I set myself on a path to solve the cube. After grueling pattern recognition and many *many* memorized algorithms, I can happily say that I know how to solve the cube. While my personal best time of just under 30 seconds is not the best in the speedcubing world, I can say that I know the cube a decent bit to speak about it.

The Rubik's Cube is a scary toy to those who do not understand it, just as Abstract Algebra is in the field of mathematics. While being able to solve the Rubik's Cube has been a useful party trick for me (and likely for others), the fact that it's a mystery to many is unfortunate. This 3x3x3 cube of (nearly) endless combinations presents a fundamental example of tangible group theory, and a way to apply methods within group theory and algebra in an otherwise

abstract way.

3.1 The Cube

Let's begin with the general cube notation, used to describe the cube (known as "Singmaster Notation"):



Let's refer to the cube's solved state as the identity. See that prime (') movements are opposite to original movements, and thus U' can also be denoted U^3 , like in D_4 . Also, see that each move has an inverse, i.e., there exists a move to return to the identity for each movement, which satisfies one of the properties for a group. We also showed the identity (solved cube), and associativity and closure are fairly trivial in this scenario. Thus, the Rubik's Cube produces a group, known as the Rubik's Cube Group.

There are, in essence, 6 real movements to describe the cube: $\{U, D, R, L, F, B\}$. Every possible state can be reached by combining these six movements. Therefore, these six moves generate the Rubik's Cube Group, and we can write $G = \langle U, D, R, L, F, B \rangle$. And just how many states does a Rubik's Cube have? A whopping 43,252,003,274,489,856,000!!

Let's break the cube down. A single 3x3x3 cube has 8 corner pieces, 12 edge pieces, and 6 center pieces. Each move of the cube permutes its smaller pieces. A face rotation moves these pieces to new positions and can also change their orientation. Thus, every move corresponds to a permutation of these pieces, and the Rubik's Cube Group can be seen as a subgroup of a permutation group acting on the set of cube pieces.

However, the cube poses certain restrictions. For example, it is impossible to swap two pieces while leaving all others fixed. This helps significantly in searching for specific algorithms; for example, a swap would be very useful but mathematically impossible, so it suffices to look for possible (albeit less efficient) algorithms.

How does this play into the Rubik's Cube and algorithms? The cube is usually solved in a sequence (by CFOP):

1. Solve the cross (edge pieces) of your choice of color. Each edge piece should correspond to the correct center piece of each color (e.x. if solving the white cross, the edge pieces [White/Red, White/Blue, White/Orange, White/Green] should match the Red, Blue, Orange, and Green center pieces, respectively.
2. Solve the corners pieces of your chosen color. Like the edges, these should correspond to the correct colors.
3. Solve the edge pieces beneath the corners.
4. Solve the opposite cross to your selected color
5. Solve the final corners

Many algorithms can be used to skip certain steps, but we will use the simplest ones. Many algorithms follow the form:

$$ABA^{-1}B^{-1}$$

Known as a commutator. This allows the desired outcome to be reached without altering too much of the rest of the cube. In the solving sequence, this is very important because it requires preserving certain areas of the cube.

1. A moves some pieces into a new region
2. B acts on those moves pieces
3. A^{-1} puts everything else back
4. B^{-1} restores most of the remaining pieces

This allows a small, controlled change. This also shows that the cube is non-abelian, as if it were abelian, $ABA^{-1}B^{-1} = AA^{-1}BB^{-1} = ee = e$.

3.1.1 Mathematical Notation

The Rubik's Cube can be written in its mathematical form: $(\mathbb{Z}_3^7 \times \mathbb{Z}_2^{11}) \rtimes ((A_8 \times A_{12}) \rtimes \mathbb{Z}_2)$. It is broken down:

- The Orientation Subgroup $(\mathbb{Z}_3^7 \times \mathbb{Z}_2^{11})$: These are the flips and twists of pieces while they remain in their slots.

1. The **Corner Orientation** $\mathbb{Z}_3^7$ are the 8 corners in 3 possible twist states. While you may assume it would be $\mathbb{Z}_3^8$, the last corner is determined by the other seven (twisting one corner is an illegal move).
 2. The **Edge Orientation** $\mathbb{Z}_2^{11}$ are the 12 edges with 2 possible orientations. Like the corners, the 12th edge is fixed by the previous 11.
- The Permutation Subgroup $(A_8 \times A_{12}) \rtimes \mathbb{Z}_2$ describes how pieces move around the cube.
 1. **Alternating Groups** $A_8 \times A_{12}$ represent the even permutations of the 8 corners and 12 edges.
 2. You cannot perform an odd permutation on just the corners or edges, you *can* perform an odd permutation on both at the same time. The $\mathbb{Z}_2$ acts as a toggle, connecting the corner and edge arrangements and ensuring both stay synchronized.
 - The Semi-Direct Product $\rtimes$ is used over $\times$ to show that the orientation of pieces is dependent on their position. If you move an edge piece from the top layer to the side layer, its available "twist" states then change relative to its new position. The Semi-Direct Product shows that these two are connected and not independent.

3.2 God's Algorithm

Interestingly enough, while the Rubik's Cube has over 43 *quintillion* states, the God's Algorithm shows that the cube, in any given state, can be solved in 20 moves or less (half-turn metric)! This conclusion follows from observing the Cayley graph of the Rubik's Cube, which shows a maximum diameter of 20. Unfortunately, God's Algorithm is only a theoretical minimum number of moves, as it requires an incredible amount of computation and is usually impossible to find for a random state of the cube. Even a quantum computer would struggle to find the optimal algorithm for each cube state, due to memory constraints. There is a proof of the algorithm, known as the superflip, which researchers discovered required 20 moves to be solved.

God's Algorithm presents an interesting topic. When I ask someone to scramble a cube, they typically want to make many moves before they are satisfied with the complexity. However, even the most complex scramble can be solved in 20 moves or less. Moreover, the superflip, a visually symmetrical scramble, is actually the most complex state a cube can be in according to God's Algorithm.

The God's Algorithm also provides a benchmark for efficiency. Humans are keen on laws, and in the cubing world, we are restricted to solving cubes within an algorithmic framework. While a world-class speedcuber might be able to

solve a cube in 50-60 moves, we will always know that there is a "perfect" solution, and that a gap exists in our understanding and true mathematical efficiency.

3.3 Closing Remarks

Abstract Algebra has helped me optimize my Rubik's Cube-solving in many ways. Like much of mathematics, I would implement algorithms by pattern recognition rather than understanding what they did. Learning group theory not only allowed me to understand the cube better, but also gave me the ability to apply certain algorithms in out-of-place situations and even create my own algorithms to meet my needs. Understanding modular arithmetic helped me realize that the U flip had only 4 possible states, and I was wasting my time flipping through each one until I recognized the pattern, rather than using a process of elimination. While this example may be a simple (and admittedly stupid) show of application, it helps highlight the thinking process that is so important for learning a topic like Abstract Algebra.

4 Conclusion

Abstract Algebra, contrary to the name, is a universal language of symmetry and structure. While it introduces abstract topics, the underlying ability is to summarize structures in general, in sweeping ways that help you understand them. From Lagrange's Theorem and Cauchy's Theorem in basic group theory, up into Cayley's Theorem, The First Fundamental Isomorphism Theorem, all the way up into Galois Theory and the Sylow Theorems (and beyond!), Abstract Algebra has brilliant ways of introducing magnificently complex topics only to generalize them in theorems with almost comedic fashion.

Abstract Algebra is often viewed as an intimidating subject (understandably so). As an undergraduate mathematics student, it was the first *real* math course I took, and the first that actually pushed me beyond studying the day before an exam. Furthermore, it was the first class that forced me to *really* understand the subject and its topics, rather than the "plug and chug" method that works so well in many of the calculus sequences and simpler topics. Until now (my second year in college), Abstract Algebra has been my most difficult course so far, yet the most enjoyable.

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